Week 8 Robust PCA

- Principal component analysis (PCA)

  n samples $\mathbb{X} = [x_1, \cdots x_n] \in \mathbb{R}^{n \times d}$

  Centered : $\mathbb{E}[X] = 0$

  minimize $\|\mathbb{X} - L\|_F$
  L : $\text{rank}(L) = r$

  Frobenius norm: $\|A\|_F = \sqrt{\sum_{ij} A_{ij}^2}$

  finds the best rank-r approximation

  suppose $\mathbb{X} = U^T \Sigma V$, $\Sigma = \begin{bmatrix} s_1 & & \\ & \ddots & \\ & & s_n \end{bmatrix}$

  solution $= U^T \begin{bmatrix} s_1 & & & \\ & \ddots & & \\ & & s_r & \\ & & & 0 \end{bmatrix} V$

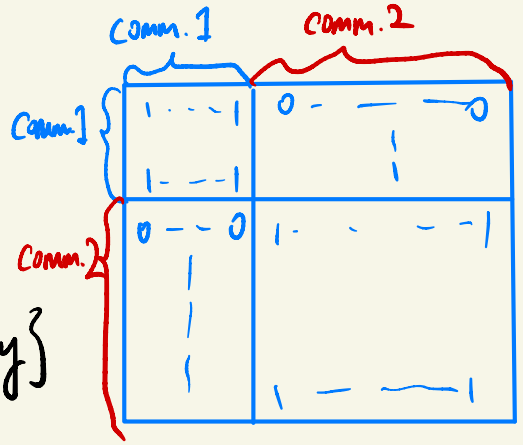
- Robust PCA

  $M = L + S \in \mathbb{R}^{m \times n}$

  L: low-rank, S: sparse

- Example : clustering / community recovery

  n persons from r communities



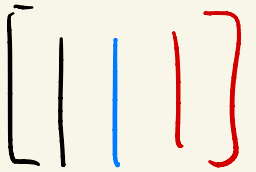
  $L_{ij} = 1\{i \text{ and } j \text{ are in the same community}\}$

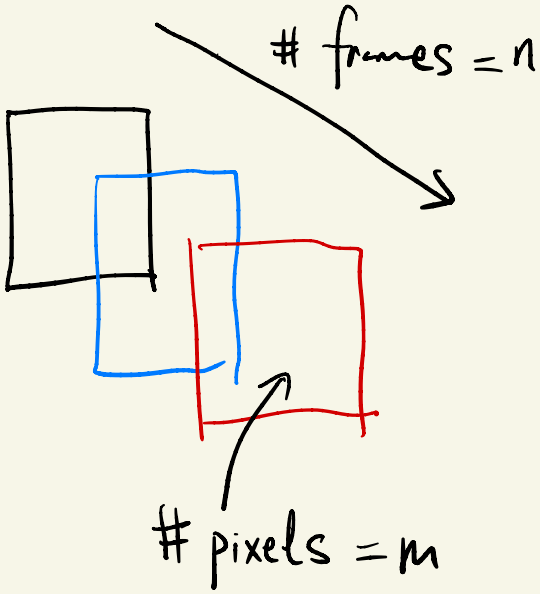
  $\text{rank}(L)$

  observe M, $S := M - L$ is sparse

- Example 2:

video data $M \in \mathbb{R}^{m \times n}$





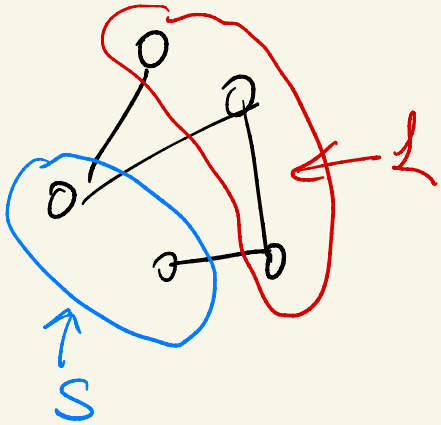
background : L low-rank

foreground : S sparse

$M = L + S$

Example : $X = [X_S, X_L]$ from a sparse Gaussian graphical model.

$$Cov(X) = \begin{bmatrix} \Sigma_S & \Sigma_{SL} \\ \Sigma_{LS} & \Sigma_L \end{bmatrix} =: \Sigma$$



$$\Sigma = \Theta^{-1}, \quad \Theta = \begin{bmatrix} \Theta_S & \Theta_{SL} \\ \Theta_{LS} & \Theta_L \end{bmatrix}$$

$$\Sigma_S^{-1} = \Theta_S - \Theta_{SL}\Theta_L^{-1}\Theta_{LS}$$

sparse

rank $\leq |L|$

Incoherence

① $M = uv^T$ $\quad u = v = [1, 0 \cdots 0]$

$$= \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & & \\ 0 & \cdots & & 0 \end{bmatrix}$$

② $u = v = \left[\frac{1}{\sqrt{n}}, \cdots \frac{1}{\sqrt{n}}\right]$

$$M = \begin{bmatrix} \frac{1}{n} & \frac{1}{n} & \cdots \\ \vdots & & \\ \cdots & & \end{bmatrix}$$ not sparse !

- Simplifying assumption $m = n$

Defn: $M = U\Sigma V^T$, $r = \text{rank}(M)$, define incoherence

$$\mu_1 = \frac{n}{r} \max\left\{ \max_i \|U^T e_i\|_2^2, \max_i \|V^T e_i\|_2^2 \right\}$$

(Standard basis)

Defn: $M = U\Sigma V^T$, define joint incoherence

$$\mu_2 := \frac{n^2}{r} \|UV^T\|_\infty^2$$

in general,

$$\mu_1 \le \mu_2 \le \mu_1^2 r$$

(see hw exercise)

$$\max_i \|V^T e_i\|_2^2$$

$$= \max_i \|UV^T e_i\|_2^2$$

$$= \max_i \| \underbrace{[UV^T]_{:,i}}_{\text{i-th column}} \|_2^2$$

$$\le \max_i \sum_\ell [UV^T]_{\ell,i}^2$$

$$\le n \max_i \max_\ell [UV^T]_{\ell,i}^2$$

$$= n \|UV^T\|_\infty^2$$

$$= \frac{\mu_2 r}{n}$$

$$\Rightarrow \frac{\mu_1 r}{n} \le \frac{\mu_2 r}{n}$$

· Algorithm:

$$\text{minimize}_{L,S} \ \|L\|_{S_1} + \lambda \|S\|_1 \quad \text{s.t. } M = L+S$$

$\|L\|_{S_1}$: 1-Schatten norm (a.k.a. nuclear norm)

Thm (CLMW11): Suppose $\text{rank}(L) \le \frac{n}{\max\{\mu_1, \mu_2\} (\log^2 n)}$

nonzero entries of S: randomly located,

$\|S\|_0 \le \rho_s n^2$, for some $\rho_s > 0$

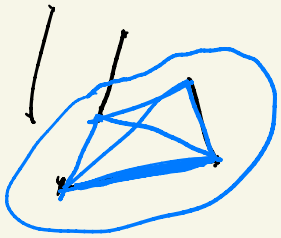
$M = L + S$.

Then Algorithm recovers L and S w.h.p.

Remark: $\max\{\mu_1, \mu_2\}$ in thm cannot be improved to μ_1, since otherwise it violates the planted clique conjecture!

Erdos Renyi graph: n vertice, there is an edge w.p. $\frac{1}{2}$

planted clique:



k-clique

$k > n^{\frac{1}{2}+\varepsilon} \Rightarrow$ efficiently algo for finding k-clique

$k < (2+\varepsilon)\log n \Rightarrow$ information-theoretically impossible

Conjecture: no efficient algo for $k < n^{\frac{1}{2}-\varepsilon}$!

- Convex geometry

setup: $M_0 = L_0 + \mathcal{U}(S_0)$

(S_0: sparse; L_0: low-rank; $\mathcal{U}$: orthogonal transform on the space of matrices (known))

Algorithm :

minimize $\|L\|_{S_1}$ s.t. $\|S\|_1 \le \|S_0\|_1$, $M_0 = L + \mathcal{U}(S)$

L_0, S_0 are optimal if no nonzero $\Delta \in \mathbb{R}^{d^2}$ s.t.

$$\exists \varepsilon > 0$$

$$\|L_0 + \varepsilon\Delta\|_{S_1} \le \|L_0\|_{S_1}$$

$$\|S_0 - \varepsilon\mathcal{U}^{-1}(\Delta)\|_1 \le \|S_0\|_1$$

$\Leftrightarrow$

$$\Delta \in \mathcal{D}(L_0, \|\cdot\|_{S_1})$$

$$\Delta = \mathcal{U}\left(-\mathcal{D}(S_0, \|\cdot\|_1)\right)$$

Previously $\delta\left(\mathcal{D}(S_0, \|\cdot\|_1)\right) = O(\|S_0\|_1 \log n)$

Thm $L_0 \in \mathbb{R}^{n^2}$, $\operatorname{rank}(L_0) = r$

$\Rightarrow$ $w\left(\mathcal{D}(L_0, \|\ \|_{S_1}) \cap B_2\right) = O(\sqrt{rn})$

(B_2: unit ball under $\|\ \|_F$)

$$\delta(\mathcal{D}(\cdots)) = O(rn)$$

Assume w.l.o.g. $L_0 = \begin{pmatrix} \Sigma & 0 \\ 0 & 0 \end{pmatrix}$ ← $\Sigma \in \mathbb{R}^{r\times r}$ diagonal

subdifferential of $\|\cdot\|_{S_1}$ at $L_0 = \begin{pmatrix} I_{r\times r} & 0 \\ 0 & A \end{pmatrix}$

↳ $\|A\|_{op} \le 1$

$$\Delta = \begin{pmatrix} \Delta_1 & \Delta_2 \\ \Delta_3 & \Delta_4 \end{pmatrix}$$

$\Delta \in$ descent cone $\Leftrightarrow \langle \Delta, L_0 \rangle \le 0$

$\Leftrightarrow \operatorname{tr}(\Delta_1) + \|\Delta_4\|_{S_1} \le 0$

$G = \begin{pmatrix} G_1 & G_2 \\ G_3 & G_4 \end{pmatrix}$ iid $N(0,1)$ entries.

$$\sup_{\Delta \in B_2 \cap D(\cdot,\cdot)} \langle G_1, \Delta_1 \rangle \le \|G_1\|_F$$

$$\sup \quad \langle G_2, \Delta_2 \rangle \le \|G_2\|_F$$

$$\text{---} \quad \le \|G_3\|_F$$

$$\sup_{\Delta_4 \in B_2 \cap D(\cdots)} \langle G_4, \Delta_4 \rangle \le \|G_4\|_{s\infty} \sup_{\Delta_4} \|\Delta_4\|_{s1}$$

$$\le \|G_4\|_{s\infty} \sup_{\Delta_4} \mathrm{tr}(-\Delta_1)$$

$$\le \|G_4\|_{s\infty} \sqrt{r}$$

$$\Rightarrow W(D(\cdots) \cap B_2)$$

$$\le \mathbb{E}[\|G_1\|_F] + \mathbb{E}\|G_2\|_F + \mathbb{E}\|G_3\|_F + \sqrt{r}\,\mathbb{E}\|G_4\|_{s\infty}$$

$$\le \sqrt{nr} + \sqrt{r} \cdot O(\sqrt{n-r})$$

$$\le O(\sqrt{nr})$$

■

Thm: If

$\lim_{n\to\infty} \frac{1}{n} \mathrm{rank}(L_0) = 0$

$\lim_{n\to\infty} \frac{\log n}{n^2} \|S_0\|_1 = 0$

then Algorithm exactly recover L_0, S_0 w. prob $\to 1$ as $n \to \infty$